

# From Social Media to Psychological Scale: An Adaptive Framework with Two-Hop Retrieval for Depression Screening

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## Abstract

Depressive disorders represent a major global public health challenge. As an increasing number of individuals share their emotional experiences and concerns on social media, researchers have shown growing interest in leveraging such data for early depression screening. However, most existing methods rely on a fixed model and a singular reasoning paradigm, which constrains their adaptability to depression detection. The limited availability of mental health-related data and variability in training data distributions across different LLMs hinder their consistent and comprehensive understanding of diverse psychological symptoms. In this paper, we propose **AdaDepression**, a framework that enables explainable depression screening through a two-hop retrieval algorithm to identify symptom-relevant posts and a two-stage adaptive routing mechanism for selecting appropriate reasoning strategies and LLMs. Specifically, we first collect representative posts from the training dataset to capture the real-world symptom expressions, and then utilize these posts to retrieve symptom-relevant posts from the user's posting history. Subsequently, we employ the Mixture of Routers (MoR), which integrates the Mixture of Experts (MoE) into the routing mechanism to select the optimal reasoning strategies and LLMs in a cascaded manner. Finally, we complete the standardized psychological questionnaire using the selected LLMs and reasoning strategies. Experimental results on the Reddit-based benchmarks demonstrate the effectiveness of the proposed method, outperforming existing studies on various metrics<sup>1</sup>.

<sup>\*</sup>This work was completed during an internship at the Institute of Artificial Intelligence, Hefei Comprehensive National Science Center.

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<sup>1</sup>Our code is released at <https://github.com/MindIntLab-HFUT/AdaDepression>.



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ACM ISBN 979-8-4007-2307-0/2026/04

<https://doi.org/10.1145/3774904.3792594>

## CCS Concepts

• **Applied computing** → **Psychology**; • **Computing methodologies** → **Natural language processing**.

## Keywords

Explainable Depression Screening, Social Media, Large Language Model, Routing Mechanism, Mixture of Experts

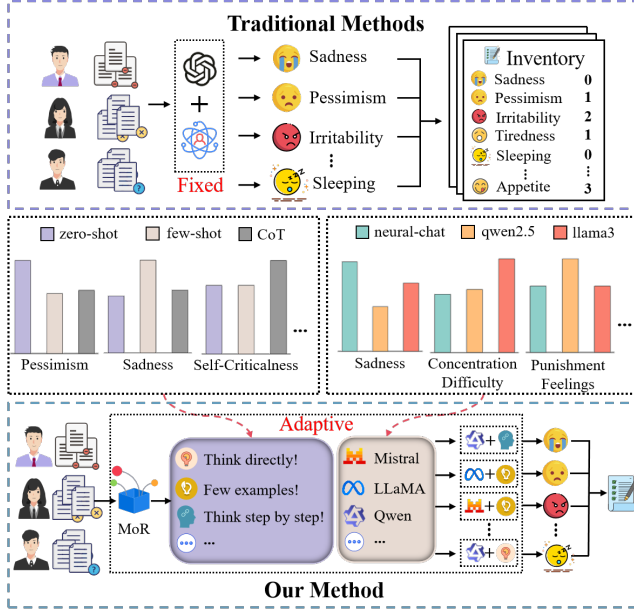
## ACM Reference Format:

Yangyang Xu, Jinpeng Hu, Peipei Song, Zhangling Duan, and Xun Yang. 2026. From Social Media to Psychological Scale: An Adaptive Framework with Two-Hop Retrieval for Depression Screening. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792594>

## 1 Introduction

According to the World Health Organization (WHO), mental health represents a significant global challenge, with depression affecting approximately 5% of the adult population worldwide. Unfortunately, due to a lack of trained health-care providers and societal stigma with mental disorders, the majority of depression cases remain undiagnosed [13]. With the growing prevalence of individuals sharing their emotions and concerns on social media platforms, the use of social media data for early depression screening has attracted increasing interest from researchers [7, 8, 20, 23, 51].

In recent years, LLMs have exhibited robust contextual understanding [19, 31, 32, 56], which has spurred a surge in studies exploring their applications in emotional reasoning [16], psychological counseling [54], and mental health, particularly the detection of depression [14, 38] and other mental health conditions [42, 57]. For example, Chen et al. [10] fine-tuned LLMs on social media data, analyzed the model's predictions and their corresponding explanations to derive insights into complex diseases. Xu et al. [53] conducted a comprehensive evaluation of LLMs across various mental health prediction tasks, including depression detection based on social media data, demonstrating a promising performance of LLMs in mental health assessment. However, these methods typically employ a black-box process that does not incorporate established



**Figure 1: A comparison of our proposed AdaDepression and other methods.**

clinical assessment tools, leading to limited explainability and clinical utility. Psychological scales serve as a vital clinical assessment tool, evaluating an individual's psychological status by measuring their scores across various symptom dimensions. Therefore, many studies have explored the integration of psychological scales into LLMs to enhance early depression screening. For instance, Aragón et al. [2] identified relevant contents, summarized this information, and utilized LLMs to complete the psychological scale. Wang et al. [47] crafted a prompt to investigate the use of LLMs in generating predictions based on the Beck Depression Inventory-II (BDI-II) [22].

Although these methods have achieved notable success, they typically utilize a fixed model architecture and a singular reasoning strategy, which lacks essential adaptability and ultimately results in suboptimal performance. As shown in Figure 1, our empirical analysis demonstrates substantial variability in predictive performance across different LLMs and reasoning strategies when applied to distinct depression symptoms, highlighting the need for adaptive selection of reasoning strategy and LLM based on the specific social media data and target depression symptom. Moreover, most existing methods retrieve symptom-related posts using idealized expressions in the scale, overlooking the linguistic gap between these expressions and user-generated content on social media platforms, which may lead to reduced retrieval effectiveness. In this paper, we propose a novel and adaptive framework that dynamically selects reasoning strategies and LLMs based on the user's posting history, enabling accurate prediction of their responses to each psychological scale symptom. Specifically, we employ a two-hop retrieval strategy that first expands symptom expressions using existing posts in the training set and then retrieves symptom-related posts based on these expanded expressions for each user. Subsequently, we encode the retrieved posts and enhance their representations with the target symptom through an attention mechanism. After

that, we introduce a two-stage routing mechanism based on the Mixture of Routers (MoR) architecture, which integrates the MoE into the routing mechanism to implement adaptive selection. The first MoR aims to select the most appropriate reasoning strategy from a predefined reasoning pool, and conditioned on this choice, the second MoR identifies the most suitable LLM from the LLM pool. In doing so, this hierarchical routing is able to select a better combination of reasoning strategy and LLM with user-specific psychological content, enhancing the adaptability and accuracy of response prediction. The entire framework is optimized using reinforcement learning, enabling effective coordination between reasoning strategies and LLMs.

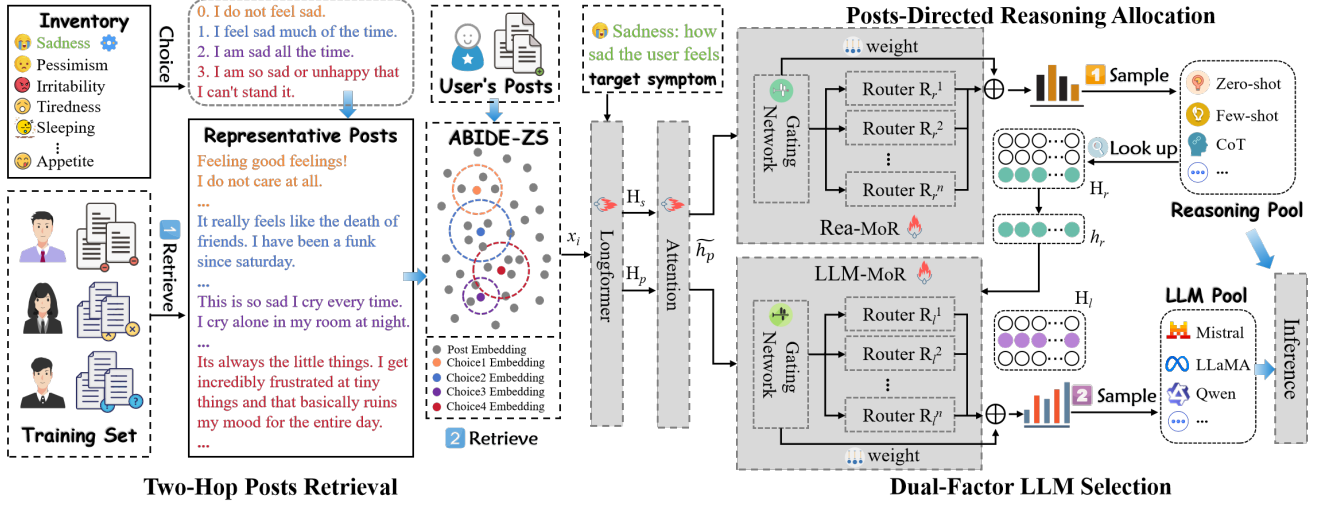
Our contributions are summarized as follows:

- This paper presents AdaDepression, a novel framework that dynamically selects appropriate LLMs and reasoning strategies based on users' posting histories to predict their scale scores.
- We employ a two-hop retrieval strategy to address the semantic gap between the expressions in the psychological scale and social media data, leading to superior retrieval performance.
- We introduce MoR, an effective architecture that integrates the MoE paradigm with a routing mechanism to enhance the selection of LLMs and their reasoning strategies.
- Experiments on the Reddit-based benchmarks show that our proposed AdaDepression achieves promising performance. Ablation studies and in-depth analysis further validate its effectiveness.

## 2 Related Work

Emotional reasoning [43] and mental health care [55, 62] have attracted mounting attention from researchers in the mental health domain. In recent years, social media has increasingly been identified as a crucial data source for analyzing mental health disorders by researchers [25, 29]. Several studies [50, 59] have shown that this type of user-generated content holds significant value in identifying depression and other mental health disorders. Consequently, computational linguistics approaches have been widely applied to establish a connection between social media text and depression [1, 9, 30, 41]. However, these black-box methods directly predict depression categories from users' writings, lacking explainability and transparency, which may undermine trust and limit their applicability in psychological practice.

To bridge this gap, many approaches have explored the integration of standardized psychological scales for explainable depression screening [61]. Moreover, CLEF eRisk 2019 [26] proposed a task named *Measuring the Severity of the Signs of Depression*, which focused on using the BDI-II to automatically predict the depression levels of social media users based on their writings. This task continued into 2020 [27] and 2021 [33]. Given the critical importance of explainability in depression screening, advanced computational methodologies have been proposed to address the task *Measuring the Severity of the Signs of Depression*. [35] explored the performance of neural language models in predicting individual depressive symptoms, based on the 21 symptoms collected in the BDI-II. Ravenda et al. [39] proposed a methodology that leverages semantic embeddings and a probabilistic neural architecture to complete the questionnaire based on social media posts automatically.



**Figure 2: The overview of our proposed AdaDepression, which consists of the following three modules: Two-Hop Posts Retrieval, Posts-Directed Reasoning Allocation, and Dual-Factor LLM Selection.**

Recently, LLMs have demonstrated growing potential for mental health assessment [11, 18, 46, 58], with several studies exploring their application to this task. For example, Wang et al. [47] meticulously designed prompts to explore the use of various LLMs in predicting BDI-II responses. More recently, [40] proposed an adaptive RAG approach to retrieve the most relevant posts and then instructed the LLM to predict the corresponding scores. However, these works often yield suboptimal outcomes, as their application of a fixed, deterministic approach to all types of content fails to account for the diversity of user-generated content and the multi-dimensional, complex nature of depression assessment. Our work differs from these studies by designing MoR for explainable depression screening, which automatically selects the optimal LLMs and reasoning strategies to enhance the predictive performance.

### 3 Methodology

#### 3.1 Overview

As illustrated in Figure 2, AdaDepression consists of the following three modules. (1) In the **Two-Hop Posts Retrieval** module, a collection of representative posts for each symptom is first identified from the training set to characterize their expressions on social media. Subsequently, these posts are employed as queries to retrieve related posts for the target user. (2) Within the **Posts-Directed Reasoning Allocation** module, we first derive a symptom-enhanced post representation by applying an attention mechanism to the retrieved posts and the corresponding symptom. This enhanced representation is then utilized to select the most suitable reasoning strategy from a reasoning pool. (3) Finally, the selection of the optimal LLM from the predefined LLM pool is performed by the **Dual-Factor LLM Selection** module, leveraging the post representation and the selected reasoning strategy, both of which are previously derived. The framework is optimized via reinforcement learning to maximize alignment with ground-truth responses, thereby enhancing overall performance.

#### 3.2 Two-Hop Posts Retrieval

The users' writing history often contains substantial content irrelevant to the target symptom, which may lead to a degradation in reasoning performance [24]. To mitigate this issue, most existing methods utilize the choices of the symptom to retrieve the relevant posts [40] (e.g., "I do not feel sad" as one of the choices for the symptom "Sadness"). However, the gap in expression patterns between social media posts and the given choices hinders retrieval effectiveness. Therefore, we proposed a two-hop retrieval algorithm to extract symptom-related content from the user's posting history. The first stage involves the collection of user-generated posts on social media from the training set, where each post represents a distinct manifestation of the symptom. For instance, as illustrated in Figure 2, posts such as "Feeling good feelings!" and "I cry alone in my room at night." are different expressions of the same symptom "Sadness". During the second stage, these representative posts, which exhibit typical social media expression patterns, are used to identify the symptom-related posts for the target user.

To be specific, in the first stage, we calculate embeddings for all posts in the training dataset using an off-the-shelf pre-trained model. Similarly, embeddings are computed for each choice corresponding to the  $i$ -th symptom using the same method. Subsequently, we employ the ABIDE-ZS method [39] to identify the most representative posts for each choice by selecting those with the top  $k^*$  cosine similarity scores. In contrast to traditional retrieval methods that rely on fixed thresholds, ABIDE-ZS dynamically optimizes  $k^*$  using the ABIDE algorithm [12]. In the second stage, we first compute the average embedding of the representative posts for each choice, forming the semantic representation of that choice. The ABIDE-ZS method is then reapplied to retrieve posts relevant to each choice for the target user. Finally, the posts corresponding to the choices of the  $i$ -th symptom are aggregated and deduplicated, yielding a refined set denoted as  $x_i$ , which is regarded as relevant to the  $i$ -th symptom.

### 3.3 Post-Directed Reasoning Allocation

In this section, we allocate appropriate reasoning strategies (e.g., Zero-shot, Few-shot, and CoT) to the subsequent inference process based on the retrieved symptom-related posts. Due to the significant diversity among these posts, a single router with limited capacity is prone to making erroneous decisions. Inspired by Zhang et al. [60], we address this challenge by integrating the MoE into the routing mechanism, designing the Mixture of Routers (MoR) architecture, as illustrated in Figure 2.

Specifically, for the  $i$ -th symptom, we first define its semantic description<sup>2</sup> (e.g., Sadness  $\rightarrow$  how sad the user feels) and then feed it into the Longformer encoder to derive its representation  $\mathbf{H}_s$  [6]. Subsequently, symptom-related posts  $x_i$  are concatenated into a single aggregated text, which is then encoded to produce the representation  $\mathbf{H}_p$  using the same method. Afterward, we integrate these representations through an attention mechanism, obtaining a symptom-enhanced representation  $\mathbf{H}_p'$  [3]. The sentence-level embedding of  $\mathbf{H}_p'$  is denoted as  $\tilde{\mathbf{h}}_p \in \mathbb{R}^{d_p}$ , corresponding to its [CLS] token representation, where  $d_p$  represents the dimensionality of the token representation.

Meanwhile, we construct a reasoning pool comprising multiple reasoning strategies, and  $\mathbf{H}_r \in \mathbb{R}^{n_r \times d_r}$  denotes their corresponding embeddings that are initialized randomly. Here,  $n_r$  is the number of reasoning strategies, and  $d_r$  is the dimensionality of each reasoning representation. We propose the MoR architecture, designed to dynamically select the most appropriate reasoning strategy from the pool. To distinguish it from the MoR used for LLM selection in the following subsection, we refer to this variant as **Rea-MoR**. In detail, the Rea-MoR consists of a gating network and  $n$  expert routers, denoted as  $R_r = \{R_r^1, R_r^2, \dots, R_r^n\}$ . Each expert router in the Rea-MoR is implemented as a simple neural network followed by a softmax function. The formula for  $R_r^i$  is as follows:

$$R_r^i(\tilde{\mathbf{h}}_p, \mathbf{H}_r) = \text{Softmax}((\tilde{\mathbf{h}}_p \mathbf{U}_r^i)(\mathbf{H}_r \mathbf{V}_r^i)^T), \quad (1)$$

where  $R_r^i(\tilde{\mathbf{h}}_p, \mathbf{H}_r) \in \mathbb{R}^{n_r}$ ,  $\mathbf{U}_r^i \in \mathbb{R}^{d_p \times d_h}$  and  $\mathbf{V}_r^i \in \mathbb{R}^{d_r \times d_h}$  are learnable parameters of  $R_r^i$ .

The symptom-enhanced post representation  $\tilde{\mathbf{h}}_p$  is first utilized to compute the scores for the  $n$  expert routers through the gating network. Then, the top  $k$  routers with the highest scores are selected to collectively determine the reasoning strategy. Each selected router utilizes the symptom-enhanced representation  $\tilde{\mathbf{h}}_p$  to compute the probability distribution over the reasoning strategies available in the reasoning pool independently. This process can be formulated as Equation 1. The final probability distribution is derived by aggregating the weighted contributions from these routers. Based on this distribution, the reasoning strategy is determined through a sampling method, with the sampled strategy representation denoted as  $\mathbf{h}_r$ , where  $\mathbf{h}_r \in \{\mathbf{H}_r^i\}_{i=1}^{n_r}$  and  $\mathbf{h}_r \in \mathbb{R}^{d_r}$ .

### 3.4 Dual-Factor LLM Selection

After determining the reasoning strategy, the next step involves selecting a suitable LLM for the  $i$ -th symptom. To this end, we first construct a pool of candidate LLMs, such as Neural-chat-7b-v3-1, Qwen2.5-32b [56], and LLaMA3-70b [45], which vary in model

size and have been demonstrated to be effective in prior studies [47]. The embeddings of these models are randomly initialized and denoted as  $\mathbf{H}_l \in \mathbb{R}^{n_l \times d_l}$ , where  $n_l$  is the number of LLMs and  $d_l$  is the dimensionality of each embedding. Subsequently, we design **LLM-MoR** consisting of a gating network and  $n$  expert routers for LLM selection, denoted as  $R_l = \{R_l^1, R_l^2, \dots, R_l^n\}$ . We posit that LLM selection is primarily influenced by two factors: the *symptom-related posts* and the *reasoning strategy* selected in the previous step. Thus, the formula for the  $i$ -th expert router  $R_l^i$  is defined as follows:

$$R_l^i(\tilde{\mathbf{h}}_p, \mathbf{h}_r, \mathbf{H}_l) = \text{Softmax}((\tilde{\mathbf{h}}_p \oplus \mathbf{h}_r) \mathbf{U}_l^i)(\mathbf{H}_l \mathbf{V}_l^i)^T), \quad (2)$$

where  $\oplus$  represent the *concat* operation,  $\mathbf{U}_l^i \in \mathbb{R}^{(d_p+d_r) \times d_h}$  and  $\mathbf{V}_l^i \in \mathbb{R}^{d_l \times d_h}$  are learnable parameters of  $R_l^i$ .

We first employ the gating network to generate a score for each expert router, and the top- $k$  routers with the highest scores are selected to jointly determine the LLM for subsequent inference. Based on the representations of symptom-related posts  $\tilde{\mathbf{h}}_p$  and the reasoning strategy  $\mathbf{h}_r$ , each selected router calculates the probability distribution over the LLMs in the predefined pool, as shown in Equation 2. The aggregation of weighted contributions from these routers yields a final probability distribution, from which the LLM is determined via a sampling method. The procedure follows an approach analogous to Post-Directed Reasoning Allocation, with the key distinction lying in the concurrent integration of two factors.

Finally, the sampled LLM and reasoning strategy are employed to predict the score of the symptom for the user. The prompt templates for different reasoning strategies are provided in Appendix A.

### 3.5 Optimization

The training loss of our AdaDepression is presented as follows:

$$\mathcal{L}_\theta = \mathbb{E}_{(x,y) \sim \mathcal{D}} [-R(\tilde{y}; x, y)], \quad (3)$$

$$R(\tilde{y}; x, y) = \begin{cases} 1 & \text{if } \tilde{y} = y, \\ -e^{|y - \tilde{y}| - 1} & \text{otherwise.} \end{cases} \quad (4)$$

where  $\mathcal{D}$  denotes the training dataset,  $x$  represents the user's symptom-related posts,  $\tilde{y}$  is the predicted score for the target symptom, and  $y$  refers to the ground truth provided by the user. We aim to achieve promising performance by maximizing the probability of generating responses that are closer to the ground truth. We apply policy gradient [49] to optimize the training loss  $\mathcal{L}_\theta$ .

Furthermore, we add an auxiliary loss  $\mathcal{L}_a$  during the training process to promote balanced utilization of routers within the MoR [15]. Therefore, the joint loss  $\mathcal{L}$  to minimize is as follows:

$$\mathcal{L} = \mathcal{L}_\theta + \alpha \mathcal{L}_a, \quad (5)$$

where  $\alpha$  is a hyperparameter to balance these two losses.

## 4 Experiments

### 4.1 Experimental Settings

**Datasets.** The datasets employed in our experiments are derived from the eRisk *Measuring the Severity of the Signs of Depression* task, spanning the 2019 [26], 2020 [28], and 2021 [34] editions. This task poses a challenge to assess the severity of depression based on users' social media posts. Researchers are provided with the complete posting history of each user and tasked with completing the

<sup>2</sup>Descriptions of all symptoms are provided in the Appendix 8.

**Table 1: Comparative results on the eRisk collections (in %). “Two-Hop” is the retrieval strategy used in our framework and the best results among all methods are highlighted in bold. DCHR\* adopts another scoring criterion: Minimal (0-13), Mild (14-19), Moderate (20-28), and Severe (29-63) [5, 48].**

| Collection  | Methods                             | Questionnaire Metrics |              |              |              | Symptom Metrics |              |
|-------------|-------------------------------------|-----------------------|--------------|--------------|--------------|-----------------|--------------|
|             |                                     | DCHR↑                 | DCHR*↑       | ADODL↑       | RMSE↓        | AHR↑            | ACR↑         |
| eRisk 2021  | DUTH_ATHENA [44]                    | 27.50                 | -            | 80.32        | -            | 33.04           | 67.86        |
|             | CYUT [52]                           | 41.25                 | -            | 83.59        | -            | 32.62           | 69.46        |
|             | UPV-Symanto [4]                     | 32.50                 | -            | 82.42        | -            | 34.17           | <b>73.17</b> |
|             | semantic-4-depression (Recall) [37] | 51.25                 | -            | 81.65        | 14.96        | 27.66           | 61.72        |
|             | semantic-4-depression (Voting) [37] | 46.25                 | -            | 81.72        | 14.83        | 27.95           | 62.40        |
|             | Wang et al. [47] (Neural-chat-7b)   | 50.00                 | 50.00        | 84.76        | 12.15        | 33.77           | 70.79        |
|             | Wang et al. [47] (Qwen2.5-32b)      | 48.75                 | 46.25        | 84.69        | 12.27        | 33.92           | 70.85        |
|             | aRAG (Qwen2.5-32b + CoT) [40]       | <b>52.50</b>          | 50.00        | 84.62        | 12.75        | 34.46           | 70.69        |
|             | aRAG (LLaMA3-70b + CoT) [40]        | 50.00                 | 46.25        | 84.74        | 12.23        | 32.38           | 70.58        |
|             | Two-Hop (LLaMA3-70b + Zero-shot)    | <b>52.50</b>          | 47.50        | 84.96        | 12.45        | 34.52           | 70.91        |
|             | Two-Hop (Neural-chat-7b + CoT)      | 47.50                 | 43.75        | 84.23        | 12.32        | 34.40           | 68.87        |
|             | Two-Hop (GPT-4o-mini + CoT)         | 51.25                 | 50.00        | 83.57        | 13.48        | 31.55           | 68.77        |
|             | Two-Hop (GPT-4o-mini + Zero-shot)   | 51.25                 | 47.50        | 82.02        | 14.46        | 33.39           | 70.40        |
|             | <b>AdaDepression(Ours)</b>          | 51.25                 | <b>51.25</b> | <b>85.85</b> | <b>11.76</b> | <b>35.60</b>    | 71.21        |
| eRisk mixed | Wang et al. [47] Neural-chat-7b     | 48.50                 | 49.25        | 83.79        | 13.11        | 33.22           | 69.81        |
|             | Wang et al. [47] Qwen2.5-32b        | 48.25                 | 47.00        | 83.65        | 13.13        | 32.94           | 70.05        |
|             | aRAG (Qwen2.5-32b + CoT) [40]       | 50.50                 | 50.25        | 83.76        | 13.23        | 33.33           | 70.24        |
|             | aRAG (LLaMA3-70b + CoT) [40]        | 47.00                 | 44.00        | 83.95        | 12.35        | 31.48           | 70.18        |
|             | Two-Hop (LLaMA3-70b + Zero-shot)    | 49.00                 | 45.00        | 83.68        | 12.36        | 33.68           | 70.78        |
|             | Two-Hop (Neural-chat-7b + CoT)      | 45.00                 | 41.00        | 83.48        | 12.90        | 32.81           | 68.18        |
|             | Two-Hop (GPT-4o-mini + CoT)         | 46.00                 | 50.00        | 82.52        | 13.94        | 31.14           | 68.40        |
|             | Two-Hop (GPT-4o-mini + Zero-shot)   | 50.00                 | 48.00        | 81.90        | 14.38        | 32.71           | 70.32        |
|             | <b>AdaDepression (Ours)</b>         | 51.25                 | <b>52.00</b> | <b>84.77</b> | <b>12.06</b> | <b>34.04</b>    | <b>70.85</b> |

standard depression questionnaire (i.e., BDI-II). The questionnaires filled out by the users (ground truth) are used to assess the quality of the predictions. The statistics of eRisk 2019, 2020 and 2021 are presented in Appendix C. The depression risk can be classified into 4 levels based on the scores of BDI-II: Minimal (0-9), Mild (10-18), Moderate (19-29), and Severe (30-63) [34]. Datasets are available for research purposes upon signing a user agreement.

In this study, we adopt the following two different dataset configurations. (1) To facilitate a fair comparison with existing work, we adhere to the evaluation framework established in the eRisk 2021 competition [34]. Specifically, the eRisk 2020 dataset is used for training, eRisk 2019 for validation, and the eRisk 2021 collection for testing. (2) To mitigate potential instability arising from the limited scale of the eRisk datasets, we utilize the eRisk 2020 dataset as the training set, while eRisk 2019 and eRisk 2021 are merged and randomly divided into five folds. In each iteration, one fold serves as the validation set, and the remaining four are used for testing. The final performance results are reported as the average across all five test runs (denoted as “eRisk mixed”).

**Baselines.** We compare our method with the following existing approaches, which can be divided into two main categories. (1) **Non-LLM-based methods:** DUTH\_ATHENA [44] generated one dense representation for each user by applying sentence transformers, and subsequently experimented with various classifiers using this representation as input. UPV-Symanto [4] involved training

a neural model on psychologist-annotated data and integrating it with a set of engineered features. In contrast, CYUT [52] focused on evaluating the performance of various RoBERTa-based classifiers. Pérez et al. [37] proposes an effective semantic pipeline for estimating depression severity in individuals based on their social media data. (2) **LLM-based methods:** Wang et al. [47] employed a fixed retrieval process with a predefined parameter  $k$ , whereas aRAG [40] introduced an adaptive retrieval mechanism that automatically determined the optimal number of samples  $k^*$  to retrieve. Additionally, we also adopted the two-hop retrieval strategy to identify relevant posts, denoted as “Two-Hop”. The LLMs employed in this study are from the LLM pool, and the prompts used across different baselines are detailed in Appendix A. For each method, we selected the LLM-prompt combinations with the two highest ADODL scores as baselines. Furthermore, to enhance the competitiveness of our baselines, we integrated GPT-4o-mini [21], a high-performance closed-source LLM, specifically for the “Two-Hop” method.

**Metrics.** We adopt the official metrics proposed in the eRisk benchmark [34] to ensure a fair comparison with competing methods. These metrics assess performance at two distinct levels. (1) At the questionnaire level, the Depression Category Hit Rate (**DCHR**) computes the percentage of depressive levels correctly estimated, while Average Difference between Overall Depression Levels (**ADODL**) measures the overall closeness between the system-filled questionnaire and the user-filled questionnaire. (2) At the symptom level,

**Table 2: Prediction performance on the psychological scale of different retrieval strategies on eRisk 2021.**

| Variants          | DCHR↑        | ADODL↑       | AHR↑         | ACR↑         |
|-------------------|--------------|--------------|--------------|--------------|
| Fixed retrieval   | 46.25        | 84.86        | 32.98        | 70.93        |
| One-Hop retrieval | 50.00        | 85.36        | 33.10        | 70.81        |
| Two-Hop retrieval | <b>51.25</b> | <b>85.85</b> | <b>35.60</b> | <b>71.21</b> |

the Average Hit Rate (AHR) represents the proportion of symptoms accurately predicted, and Average Closeness Rate (ACR) assesses the average closeness between the predicted and actual scores for each symptom. Additionally, we introduce the Root-Mean-Square Error (RMSE) to assess the models' predictions of the BDI-II score. **Implementation Details.** In the implementation of our proposed Adadepression, we utilize *all-MiniLM-L12-v2*<sup>3</sup> to calculate the embeddings for all the posts and each choice of symptoms. Rea-MoR and LLM-MoR share two hyperparameters: the maximum number of routers  $n$ , and the number of selected routers  $k$ , with  $n$  set to 8 and  $k$  set to 2<sup>4</sup>. The reasoning pool consists of **Zero-shot**, **Few-shot**, and **CoT**, while the LLM pool includes **Neural-chat-7b-v3-1**<sup>5</sup>, **Qwen2.5-32b**, and **LLaMA3-70b**, all derived from Ollama, an open-source tool for managing and deploying LLMs. We set the temperature to 0 during inference and configured the hyperparameter  $\alpha$  to 0.05 during the training process. The learning rate is established at 0.001, and the maximum number of training epochs is set to 10. All experiments are conducted on an A800 GPU.

## 4.2 Comparison with Existing Methods

As shown in Table 1, the experimental results reveal the following key findings. First, we find that non-LLM-based methods, such as UPV-Symanto and UNSL, achieve competitive performance on the symptom-level metrics. This performance may stem from their targeted training at the symptom level. However, this advantage fails to generalize to questionnaire-level assessment, leading to limited improvements in assessing users' overall depression levels. Besides, our proposed method outperforms other LLM-based approaches on almost all the metrics, highlighting its effectiveness in explainable depression screening. This improvement may be attributed to the following contributions. On the one hand, the two-hop retrieval process improves the relevance of retrieved posts, which in turn enhances the LLM's understanding of the user's overall depression level. On the other hand, the adaptive mechanism customizes the selection of LLMs and reasoning strategies to align with each user's specific posts and target symptoms, thereby boosting overall predictive performance. Furthermore, we find that on the ADODL metric, which is recognized as the most critical metric for measuring depression at the population level [47], the method proposed by Wang et al. [47] and aRAG [40] slightly surpass the "Two-Hop" method, yet remain significantly inferior to our proposed method, which also employs a two-hop retrieval strategy. This performance advantage arises from our adaptive selection mechanism, which leverages diverse reasoning strategies to address different queries,

**Table 3: Retrieval performance of different strategies on eRisk 2019. The best scores are in bold.**

| Strategy          | Avg. Posts | Precision   | Recall      | F1          |
|-------------------|------------|-------------|-------------|-------------|
| Fixed retrieval   | 147.4      | 0.09        | 0.61        | 0.16        |
| One-Hop retrieval | 177.5      | 0.08        | 0.64        | 0.14        |
| Two-Hop retrieval | 142.6      | <b>0.10</b> | <b>0.67</b> | <b>0.18</b> |

**Table 4: Prediction performance on the psychological scale of different routing mechanisms on eRisk 2021.**

| Variants                 | DCHR↑        | ADODL↑       | AHR↑         | ACR↑         |
|--------------------------|--------------|--------------|--------------|--------------|
| Random routing           | 43.75        | 81.61        | 33.39        | 69.31        |
| Single router            | 46.25        | 84.25        | 33.51        | 70.36        |
| w/o $\mathcal{L}_\alpha$ | 50.00        | 84.68        | 33.69        | 71.09        |
| MoR                      | <b>51.25</b> | <b>85.85</b> | <b>35.60</b> | <b>71.21</b> |

providing an alternative to manually designed high-quality prompts that typically demand considerable time and effort.

## 4.3 Retrieval Strategy Analysis

In this section, we present a comparative analysis of the results derived from different retrieval strategies. (1) **Fixed retrieval** retrieves top  $k$  posts where  $k$  is fixed and predefined ( $k=10$ ); (2) **One-Hop retrieval** utilizes the adaptive retrieval algorithm ABIDE-ZS to retrieve posts based on the symptom choices without expanding their expressions; (3) **Two-Hop retrieval** refers to the strategy employed in our proposed framework. As shown in Table 2, the Two-Hop retrieval strategy yields superior prediction performance on the psychological scale compared to both Fixed and One-Hop retrieval strategies, demonstrating its effectiveness in facilitating more accurate predictions of users' psychological scale scores.

To further investigate the mechanisms responsible for the performance improvement of the Two-Hop retrieval strategy, we compare and analyze the retrieval performance of different strategies on the eRisk 2019 dataset, as shown in Table 3. The analysis employs BDI-Sen [36], a dataset built upon the eRisk 2019 collections, where each sentence has been manually annotated with binary relevance labels. We posit that posts containing at least one sentence labeled as relevant (i.e., with a label of 1) in the BDI-Sen dataset as the retrieval targets. We find that the Two-Hop retrieval strategy not only achieves the highest Recall with the fewest retrieved posts on average, but also attains the best performance in terms of both Precision and F1 score, surpassing all other retrieval strategies. This highlights the effectiveness of our Two-Hop retrieval strategy in mitigating the gap in expression patterns between symptoms in the BDI-II and user-generated content on social media, thereby improving the retrieval performance.

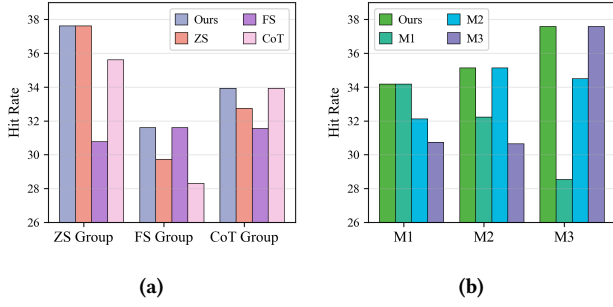
## 4.4 Routing Mechanism Analysis

To evaluate the effectiveness of the routing mechanism in our framework, we compare the following variants against the proposed MoR architecture. (1) **Single router**: a variant using a single router instead of the MoR architecture; (2) **Random routing** randomly selects a reasoning strategy and an LLM for each query; (3) **w/o  $\mathcal{L}_\alpha$**  removes the auxiliary loss  $\mathcal{L}_\alpha$  during the training phase.

<sup>3</sup><https://huggingface.co/sentence-transformers/all-MiniLM-L12-v2>

<sup>4</sup>The experiments related to the hyperparameters can be found in the Appendix 6.

<sup>5</sup>[https://ollama.com/fakezeta/neural-chat-7b-v3-1:Q5\\_K\\_M](https://ollama.com/fakezeta/neural-chat-7b-v3-1:Q5_K_M)



**Figure 3: Results of our proposed MoR on different groups.** In Figure 3a, we classify the samples in eRisk 2021 into three groups based on the reasoning strategy employed by our framework. Figure 3b presents the samples divided into three groups according to the LLM selected in our framework.

As shown in Table 4, we can obtain several observations from the results. Firstly, it is evident that our proposed MoR achieves a superior performance compared to the Single Router, which can be attributed to its efficacy in leveraging the advantages of MoE to handle diverse inputs and consequently augment the model’s overall capacity. Moreover, the proposed MoR achieves substantial improvements over Random routing, demonstrating its capacity to meaningfully guide the selection of both reasoning strategy and LLM through reinforcement learning. Finally, we find that incorporating  $\mathcal{L}_\alpha$  further enhances the model’s performance. This improvement likely stems from  $\mathcal{L}_\alpha$  promoting balanced training across routers, thereby boosting the overall predictive performance.

Furthermore, we conduct a series of experiments on the eRisk 2021 collection to explore the mechanism underlying the performance improvement of our proposed MoR, as shown in Figure 3. Specifically, the test samples from the eRisk 2021 collection are categorized in two ways. First, they are grouped by the reasoning strategy employed by our framework, which includes ZS Group (Zero-shot), FS Group (Few-shot), and CoT Group (CoT), as shown in Figure 3a. For instance, samples in the ZS Group indicate that the Zero-shot strategy is selected as the final reasoning approach in our proposed AdaDepression. Second, they are categorized by the LLM used, namely M1 (Neural-chat-7b-v3-1), M2 (Qwen2.5-32b), or M3 (LLaMA3-70b), as depicted in Figure 3b. For example, samples in the M1 Group correspond to instances where Neural-chat-7b-v3-1 is chosen as the final LLM in our framework. For each group, the performance of our method is compared against the two alternative approaches. Experimental results confirm the superiority of our method over alternative approaches across all groups. For example, within the ZS Group, the AdaDepression framework’s Zero-Shot strategy significantly outperforms both Few-Shot and Chain-of-Thought (CoT) reasoning in hit rate. This consistent advantage across the FS and CoT Groups validates the effectiveness of our reasoning strategy routing. Similarly, the results from the LLM-based groups affirm the robustness of our LLM selection mechanism. These findings collectively demonstrate that the routing mechanism of our proposed MoR successfully tailors the choice of LLM and reasoning strategy to the specific characteristics of the input data and the target depression symptom.

**Table 5: Effect of different reasoning and LLM pools on the performance of our proposed method.** The abbreviations ZS and FS denote Zero-shot and Few-shot, respectively. M1, M2, and M3 represent the LLMs Neural-chat-7b-v3-1, Qwen2.5-32b, and LLaMA3-70b, respectively.

| Reasoning Pool |    |     |        |       | LLM Pool |    |    |        |       |
|----------------|----|-----|--------|-------|----------|----|----|--------|-------|
| ZS             | FS | CoT | ADODL↑ | RMSE↓ | M1       | M2 | M3 | ADODL↑ | RMSE↓ |
| ✓              | ✗  | ✗   | 84.62  | 12.38 | ✓        | ✗  | ✗  | 84.72  | 12.59 |
| ✗              | ✓  | ✗   | 84.13  | 12.95 | ✗        | ✓  | ✗  | 84.34  | 12.92 |
| ✗              | ✗  | ✓   | 84.68  | 12.63 | ✗        | ✗  | ✓  | 84.46  | 12.65 |
| ✓              | ✓  | ✗   | 84.79  | 12.31 | ✓        | ✓  | ✗  | 84.92  | 12.46 |
| ✓              | ✗  | ✓   | 85.04  | 12.15 | ✓        | ✗  | ✓  | 85.06  | 12.28 |
| ✗              | ✓  | ✓   | 84.86  | 12.35 | ✗        | ✓  | ✓  | 84.90  | 12.17 |
| ✓              | ✓  | ✓   | 85.85  | 11.76 | ✓        | ✓  | ✓  | 85.85  | 11.76 |

#### 4.5 Effect of the Reasoning and LLM Pools

As shown in Table 5, we conduct a series of experiments by varying the elements within the pools and draw the following conclusions. First, the correlation between performance improvements and increased diversity in both reasoning and LLM pools further validates the rationale and effectiveness of our adaptive selection mechanism, a phenomenon primarily attributed to the reinforcement learning-based optimization of our proposed framework that enables it to select optimal reasoning strategies and LLMs. Second, we find that when using a single reasoning strategy, the performance of *Few-shot* is relatively poor, and removing it from the reasoning pool results in only a minor decrease in performance. This discrepancy may arise from the high similarity among posts with different scores in the examples, which makes it difficult for the LLMs to distinguish the intensity of different scores, ultimately contributing to the observed performance decline. Third, *Neural-chat-7b-v3-1* performs quite well within the LLM pool, demonstrating that small models can also offer significant value for this task.

#### 4.6 Case Study

In Figure 4, we present an example from the eRisk 2021 dataset under the symptom “Pessimism” to demonstrate how our proposed AdaDepression works. During the retrieval phase, our Two-Hop retrieval strategy captures more comprehensive content, including not only expressions reflecting negative aspects (e.g., “Everyone thinks I want to commit suicide”) but also those reflecting positive aspects (e.g., “Nah life is good”). In contrast, the One-Hop retrieval strategy fails to retrieve those positive expressions, which leads to an overestimated prediction score (score=3) when using the same inference mode, whereas our method produces an accurate prediction (score=1). Moreover, the use of MoR allows our approach to select a more appropriate reasoning strategy along with a suitable LLM, with Qwen in a Zero-shot setting being our specific choice. Compared to other inference configurations, including LLaMA with Zero-shot and Qwen with CoT, our approach not only achieves a more accurate prediction score for this specific symptom but also yields superior performance in both Total Score and Severity Level predictions, highlighting the effectiveness of the proposed method.

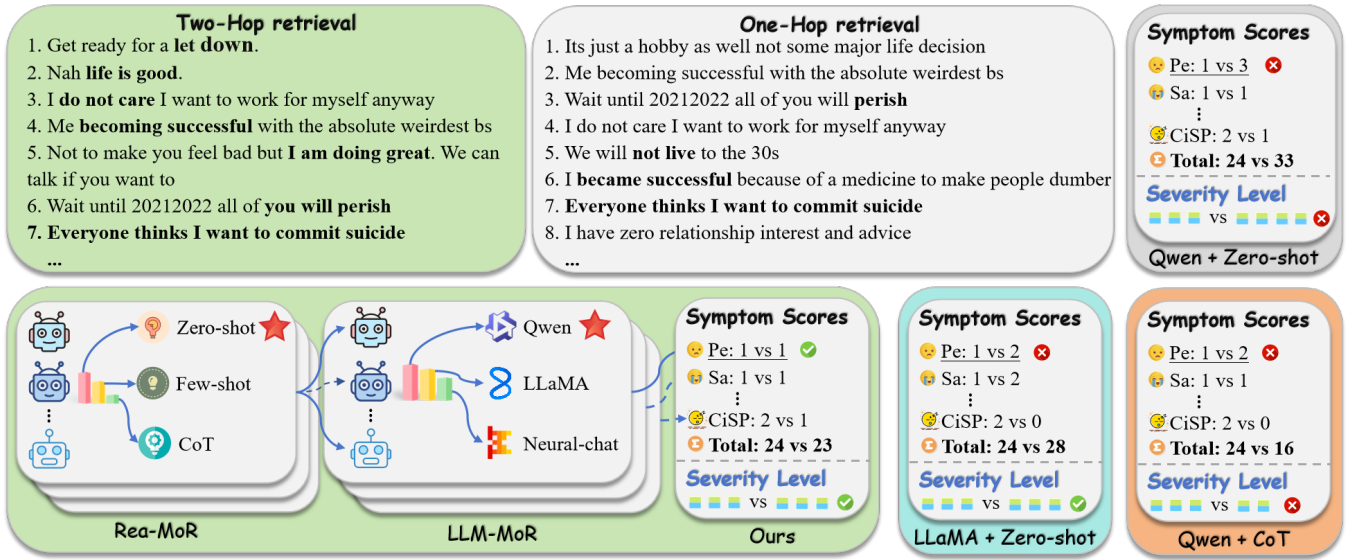


Figure 4: Case study on the symptom Pessimism. The green background represents our method, while the gray, blue, and orange backgrounds denote other methods. For clarity and brevity, single-word symptoms are represented by their first two letters, while multi-word symptoms use the first letter of each word.

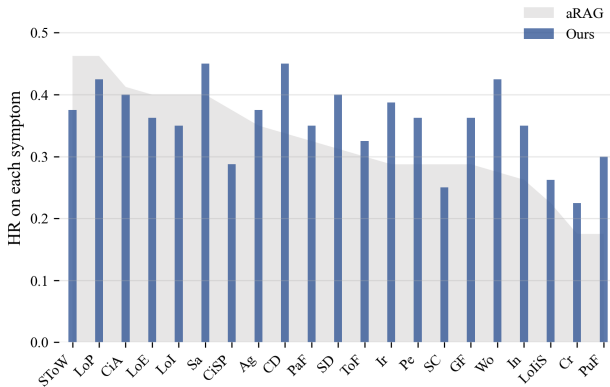


Figure 5: Further analysis at the symptom level. Symptom abbreviations are the same as those in Figure 4.

#### 4.7 Fine-grained Analysis on Symptom

Figure 5 illustrates the Hit Rate (HR) performance on each symptom of BDI-II for our method versus the aRAG baseline implemented with LLaMA3-70b and CoT. It is evident that our proposed method outperforms aRAG on 14 out of the 21 symptoms, with significant improvements notably observed in Worthlessness (Wo), Irritability (Ir), Concentration Difficulty (CD), and Punishment Feelings (PuF). These results demonstrate the superior performance of our method over the aRAG baseline, highlighting a clear advantage of our two-hop retrieval strategy and adaptive selection mechanism. Moreover, we find that both our method and aRAG achieve strong performance on Loss of Pleasure (LoP), Sadness (Sa), and Changes in Appetite (CiA), with HR scores surpassing 0.4. In contrast, their performance is comparatively lower on Self-Criticalness (SC), Loss of Interest in Sex (LoIS), and Crying (Cr), where the HR values remain below

0.3. This may be because LLMs demonstrate a stronger grasp of straightforward symptoms, such as Sadness (Sa), while showing limited capacity to interpret more nuanced symptoms like Self-Criticalness (SC). Furthermore, this pattern could also stem from the fact that symptoms such as Changes in Appetite (CiA) are more frequently discussed by users on social media platforms, in contrast to topics like Loss of Interest in Sex (LoIS), which receive comparatively less attention.

## 5 Conclusion

In this paper, we propose AdaDepression, a novel and adaptive framework to predict users' responses to each symptom of the psychological scale based on their posting history. Given the variability in predictive performance across different LLMs and reasoning strategies for the same symptom, we introduce a two-stage adaptive routing mechanism that dynamically selects the most suitable LLM and reasoning strategy based on the specific social media data and target symptom. Moreover, we introduce a two-hop retrieval process, utilizing the real-world expressions of symptoms to search relevant posts, thereby mitigating the linguistic gap between the questionnaire symptoms and user-generated content. Experimental results on eRisk collections demonstrate that our proposed framework achieves promising performance, indicating its effectiveness for explainable depression screening.

## 6 Acknowledgments

This work was supported by the National Natural Science Foundation of China (NSFC) under Grants U22A2094, 62402158, and 62402471, the Anhui Provincial Key Research and Development Project (202304a05020068), and the advanced computing resources provided by the Supercomputing Center of the USTC.

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## A Prompt Templates

This section presents the prompt templates for AdaDepression and the LLM-based baselines, shown in Table 6 and Table 7, respectively.

**Table 6: Prompt templates for our proposed AdaDepression.**

| Zero-shot Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>You are a professional psychologist. Below are the user’s REDDIT POSTS. Based on the content of these posts and the scoring criteria, predict the user’s score (0–3) on the given BDI-II item, and provide a clear explanation.</p> <p>### REDDIT POSTS ###</p> <p>{REDDIT_POSTS}</p> <p>### BDI-II ITEM ###</p> <p>{ITEM}</p> <p>### SCORING CRITERIA ###</p> <p>{choices}</p> <p>The predicted score must be an integer ranging from 0 to 3. Your response must strictly follow the output format without any other words or content:</p> <p>Score: [score]</p> <p>Explanation: [explanation]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Few-shot Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p>You are a professional psychologist. Below are the user’s REDDIT POSTS. Based on the content of these posts and the scoring criteria, predict the user’s score (0–3) on the given BDI-II item.</p> <p>### BDI-II ITEM ###</p> <p>{ITEM}</p> <p>### SCORING CRITERIA ###</p> <p>{choices}</p> <p>You can refer to the following examples.</p> <p>### EXAMPLES ###</p> <p>{examples}</p> <p>### TARGET ###</p> <p>### REDDIT POSTS ###</p> <p>{REDDIT_POSTS}</p> <p>The predicted score must be an integer ranging from 0 to 3. Your response must strictly follow the output format without any other words or content:</p> <p>Score: [score]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| CoT Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>You are a professional psychologist. Based on the user’s REDDIT POSTS provided below, determine their score (0–3) on the specified BDI-II item according to the scoring criteria.</p> <p>### REDDIT POSTS ###</p> <p>{REDDIT_POSTS}</p> <p>### BDI-II ITEM ###</p> <p>{ITEM}</p> <p>### SCORING CRITERIA ###</p> <p>{choices}</p> <p>Follow these steps:</p> <p>Step 1: Identify posts that contain first-person statements related to the symptom described in the BDI-II item. Output them as a list of strings.</p> <p>Step 2: Based on the content of these relevant posts and the scoring criteria, determine the user’s score (0–3) for the given BDI-II item.</p> <p>Step 3: Generate an explanation that corresponds to this score.</p> <p>Your explanation must be based solely on the identified posts.</p> <p>The predicted score must be an integer ranging from 0 to 3. Your response must strictly follow the output format without any other words or content:</p> <p>** Step 1 Output **</p> <p>Identified Posts: [“post 1”, “post 2”, ... “post n”]</p> <p>** Step 2 Output **</p> <p>Score: [score]</p> <p>** Step 3 Output **</p> <p>Explanation: [explanation]</p> |

**Table 7: Prompt templates for the LLM-based baselines.**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>aRAG [40] System Message</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| You are a psychological assistant specializing in administering and interpreting standardized psychological assessments. Your task is to assist in filling out the Beck Depression Inventory-II (BDI-II), a widely used self-report questionnaire designed to assess the severity of depression, based on the user's Reddit Posts. Determine the most appropriate answer to each questionnaire item. Please provide only a score from 0 to 3.                                                                                                                                                             |
| <b>aRAG [40] Direct</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Consider the following Reddit post: [POSTS]. These posts are the most relevant posts from the user according to the item: [ITEM]. Which of the following choices do you think is the most appropriate response: [CHOICES]. Report only the final score, only one value between 0 and 3 included based on the intensity of [ITEM].                                                                                                                                                                                                                                                                         |
| <b>aRAG [40] CoT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Follow the instructions. 1. Consider the following Reddit posts: [POSTS]. 2. Identify which posts are the most relevant for answering a question related to [ITEM]. 3. Based on the relevant Reddit posts, choose which of the following choices seems most appropriate as a response: [CHOICES]. Why does this choice stand out as the best match given the user's current psychological state? Think about the reasoning behind this choice step by step. 4. Finally, report the final score (0-3) based on the intensity of [ITEM]. Use the reasoning from the previous steps to justify your scoring. |
| <b>Wang et al. [47]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| ### Instruction: The following paragraph was concatenated from a user's posts on social media. Suppose you are a psychiatrist who prefer to give minor diagnoses rather than serious ones, read the posts as a whole, determine the level of "[rephrased symptom]" and give a number in 0, 1, 2 or 3, then explain why.                                                                                                                                                                                                                                                                                   |
| ### User's posts: [input text]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| ### Level (0, 1, 2 or 3):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Table 8: Descriptions for each symptom.**

| Symptom                     | Description                                |
|-----------------------------|--------------------------------------------|
| Sadness                     | how sad you feel                           |
| Pessimism                   | how discouraged you feel about the future  |
| Past Failure                | how much you feel like a failure           |
| Loss of Pleasure            | how much pleasure you have lost in things  |
| Guilty Feelings             | how often you feel guilty                  |
| Punishment Feelings         | how much you feel punished                 |
| Self-Dislike                | how much you feel disappointed in yourself |
| Self-Criticalness           | how often you criticize or blame yourself  |
| Suicidal Thoughts or Wishes | how often you think about harming yourself |
| Crying                      | how often you cry                          |
| Agitation                   | how restless or agitated you feel          |
| Loss of Interest            | how much interest you have lost in things  |
| Indecisiveness              | how difficult it is to make decisions      |
| Worthlessness               | how much you feel worthless                |
| Loss of Energy              | how much energy you have lost              |
| Changes in Sleeping Pattern | how much your sleep patterns have changed  |
| Irritability                | how irritable you feel                     |
| Changes in Appetite         | how much your appetite has changed         |
| Concentration Difficulty    | how difficult it is for you to concentrate |
| Tiredness or Fatigue        | how tired or fatigued you feel             |
| Loss of Interest in Sex     | how much interest you have lost in sex     |

## B Descriptions for Each Symptom

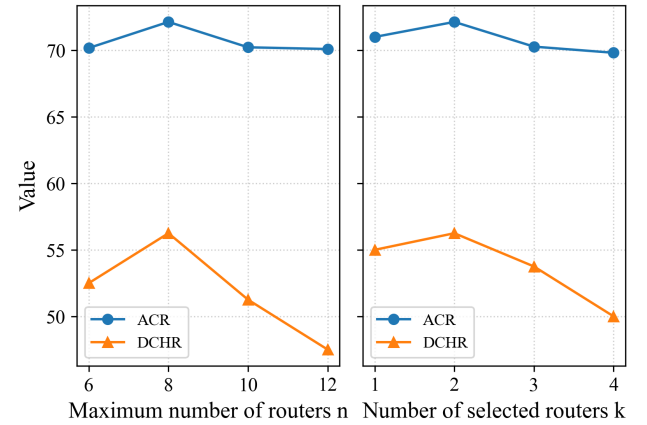
The mapping relationship from symptom to its description is provided in Table 8.

## C Statistics of the eRisk collections

The statistics of eRisk 2019, 2020, and 2021 are presented in Table 9.

**Table 9: Statistics of the eRisk collections. Min., Mil., Mod., and Sev. denote the four depression risk levels based on BDI-II scores: Minimal (0–9), Mild (10–18), Moderate (19–29), and Severe (30–63), respectively.**

| Collections | Severity Level |      |      |      | Total | Avg. Post per User |
|-------------|----------------|------|------|------|-------|--------------------|
|             | Min.           | Mil. | Mod. | Sev. |       |                    |
| eRisk 2019  | 4              | 4    | 4    | 8    | 20    | 520.55             |
| eRisk 2020  | 10             | 23   | 18   | 19   | 70    | 475.43             |
| eRisk 2021  | 6              | 13   | 27   | 34   | 80    | 388.06             |



**Figure 6: Effect of hyperparameters in MoR architecture.**

## D Studies on Hyperparameters in MoR

In this study, the maximum number of routers  $n$  and the number of selected routers  $k$  are shared between the Rea-MoR and LLM-MoR. We conduct a series of experiments to identify the optimal values. Figure 6 (left) illustrates the influence of the maximum number of routers  $n$  on performance with  $k = 2$ , which indicates that the performance initially improves with increasing  $n$ , reaches an optimal value at  $n = 8$ , and subsequently declines as  $n$  continues to rise. Figure 6 (right) represents the impact of the number of selected routers  $k$  on performance with  $n$  fixed at 8, exhibiting a similar trend where optimal performance is achieved at  $k = 2$ . Based on these findings, we set the maximum number of routers  $n$  to 8 and the number of selected routers  $k$  to 2 in this work.

## E Beck Depression Inventory-II

The Beck Depression Inventory-II (BDI-II) is a widely used, 21-item self-report questionnaire designed to assess the severity of depressive symptoms in individuals, aligning with the diagnostic criteria for major depressive disorder outlined in the DSM-IV [17]. It evaluates emotional, cognitive, motivational, and physiological symptoms such as sadness, pessimism, and fatigue, with respondents rating the frequency of each symptom over the past two weeks on a 4-point scale (0 to 3). Total scores range from 0 to 63, with higher scores indicating more severe depressive symptoms, and it is commonly utilized in clinical, research, and educational settings to aid in diagnosis, treatment planning, and symptom monitoring. Figure 7 and 8 show the content of BDI-II.

| Beck Depression Inventory-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Sadness</b></p> <p>0. I do not feel sad.</p> <p>1. I feel sad much of the time.</p> <p>2. I am sad all the time.</p> <p>3. I am so sad or unhappy that I can't stand it.</p> <p><b>Pessimism</b></p> <p>0. I am not discouraged about my future.</p> <p>1. I feel more discouraged about my future than I used to be.</p> <p>2. I do not expect things to work out for me.</p> <p>3. I feel my future is hopeless and will only get worse.</p> <p><b>Past Failure</b></p> <p>0. I do not feel like a failure.</p> <p>1. I have failed more than I should have.</p> <p>2. As I look back, I see a lot of failures.</p> <p>3. I feel I am a total failure as a person.</p> <p><b>Loss of Pleasure</b></p> <p>0. I get as much pleasure as I ever did from the things I enjoy.</p> <p>1. I don't enjoy things as much as I used to.</p> <p>2. I get very little pleasure from the things I used to enjoy.</p> <p>3. I can't get any pleasure from the things I used to enjoy.</p> <p><b>Guilty Feelings</b></p> <p>0. I don't feel particularly guilty.</p> <p>1. I feel guilty over many things I have done or should have done.</p> <p>2. I feel quite guilty most of the time.</p> <p>3. I feel guilty all of the time.</p> <p><b>Punishment Feelings</b></p> <p>0. I don't feel I am being punished.</p> <p>1. I feel I may be punished.</p> <p>2. I expect to be punished.</p> <p>3. I feel I am being punished.</p> <p><b>Self-Dislike</b></p> <p>0. I feel the same about myself as ever.</p> <p>1. I have lost confidence in myself.</p> <p>2. I am disappointed in myself.</p> <p>3. I dislike myself.</p> <p><b>Self-Criticalness</b></p> <p>0. I don't criticize or blame myself more than usual.</p> <p>1. I am more critical of myself than I used to be.</p> <p>2. I criticize myself for all of my faults.</p> <p>3. I blame myself for everything bad that happens.</p> <p><b>Suicidal Thoughts or Wishes</b></p> <p>0. I don't have any thoughts of killing myself.</p> <p>1. I have thoughts of killing myself, but I would not carry them out.</p> <p>2. I would like to kill myself.</p> <p>3. I would kill myself if I had the chance.</p> <p><b>Crying</b></p> <p>0. I don't cry anymore than I used to.</p> <p>1. I cry more than I used to.</p> <p>2. I cry over every little thing.</p> <p>3. I feel like crying, but I can't.</p> <p><b>Agitation</b></p> <p>0. I am no more restless or wound up than usual.</p> <p>1. I feel more restless or wound up than usual.</p> <p>2. I am so restless or agitated that it's hard to stay still.</p> <p>3. I am so restless or agitated that I have to keep moving or doing something.</p> |

Figure 7: Beck Depression Inventory-II (Part 1)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Loss of Interest</b></p> <p>0. I have not lost interest in other people or activities.</p> <p>1. I am less interested in other people or things than before.</p> <p>2. I have lost most of my interest in other people or things.</p> <p>3. It's hard to get interested in anything.</p> <p><b>Indecisiveness</b></p> <p>0. I make decisions about as well as ever.</p> <p>1. I find it more difficult to make decisions than usual.</p> <p>2. I have much greater difficulty in making decisions than I used to.</p> <p>3. I have trouble making any decisions.</p> <p><b>Worthlessness</b></p> <p>0. I do not feel I am worthless.</p> <p>1. I don't consider myself as worthwhile and useful as I used to.</p> <p>2. I feel more worthless as compared to other people.</p> <p>3. I feel utterly worthless.</p> <p><b>Loss of Energy</b></p> <p>0. I have as much energy as ever.</p> <p>1. I have less energy than I used to have.</p> <p>2. I don't have enough energy to do very much.</p> <p>3. I don't have enough energy to do anything.</p> <p><b>Changes in Sleeping Pattern</b></p> <p>0. I have not experienced any change in my sleeping pattern.</p> <p>1a. I sleep somewhat more than usual.</p> <p>1b. I sleep somewhat less than usual.</p> <p>2a. I sleep a lot more than usual.</p> <p>2b. I sleep a lot less than usual.</p> <p>3a. I sleep most of the day.</p> <p>3b. I wake up 1-2 hours early and can't get back to sleep.</p> <p><b>Irritability</b></p> <p>0. I am no more irritable than usual.</p> <p>1. I am more irritable than usual.</p> <p>2. I am much more irritable than usual.</p> <p>3. I am irritable all the time.</p> <p><b>Changes in Appetite</b></p> <p>0. I have not experienced any change in my appetite.</p> <p>1a. My appetite is somewhat less than usual.</p> <p>1b. My appetite is somewhat greater than usual.</p> <p>2a. My appetite is much less than before.</p> <p>2b. My appetite is much greater than usual.</p> <p>3a. I have no appetite at all.</p> <p>3b. I crave food all the time.</p> <p><b>Concentration Difficulty</b></p> <p>0. I can concentrate as well as ever.</p> <p>1. I can't concentrate as well as usual.</p> <p>2. It's hard to keep my mind on anything for very long.</p> <p>3. I find I can't concentrate on anything.</p> <p><b>Tiredness or Fatigue</b></p> <p>0. I am no more tired or fatigued than usual.</p> <p>1. I get more tired or fatigued more easily than usual.</p> <p>2. I am too tired or fatigued to do a lot of the things I used to do.</p> <p>3. I am too tired or fatigued to do most of the things I used to do.</p> <p><b>Loss of Interest in Sex</b></p> <p>0. I have not noticed any recent change in my interest in sex.</p> <p>1. I am less interested in sex than I used to be.</p> <p>2. I am much less interested in sex now.</p> <p>3. I have lost interest in sex completely.</p> |
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Figure 8: Beck Depression Inventory-II (Part 2)